RoO Input–Output Codes for PTA 6–10: Results

31 October 2026 · August Liu. Companion to the coding workflow document (general coding rules, 31 October 2026) and the variable explanation file.

1. Data

- Five agreements: Turkey–Singapore (HS2012), Nicaragua–Taiwan, Chile–Korea, Oman–US and Canada–Chile (all HS2002). Each row is one (restricted input, output) pair at HS6 from the agreement's product-specific rules: 1,171,051 rows in all. The 2,273 Turkey–Singapore rows whose input is "WO" (wholly obtained) or "any" rather than an HS code were deleted (decision of 28 October).
- One Stata file per agreement (*<PTA>_coded_2026_10_31.dta*) with 36 variables; Turkey–Singapore has one more (special_rule, from its original file).
- All pairs are coded on HS2002. Turkey–Singapore was mapped from HS2012 with the UN correlation table, and 2,390 of its rows map to several HS2002 pairs. 591 rows, where two HS2012 codes fall into the same HS2002 code, were coded on the HS2012 codes by the two AI coders (413 of them are among the 2,390). The other multi-mapped rows take the highest DV among their HS2002 pairs, with uncertain = 1 if the pairs differ, except self-pairs (the same HS2012 code), which are locked as the same good.
- The five agreements contain 405,004 unique HS2002 pairs. 349,143 of them (86.2%) were already coded in PTA 1–5 and were copied; the other 55,861 were coded under the workflow. A pair has the same code in every agreement.

2. Main results

Table 1. Input–output links by agreement

Agreement	Rows	DV = 1	Share	DV_strict = 1	DV_excl_remelt = 1	Uncertain	Uncertain / DV = 1
Turkey–Singapore	21,873	14,559	66.6%	9,018	14,436	5,541	38.1%
Nicaragua–Taiwan	211,027	139,179	66.0%	120,721	138,150	18,458	13.3%
Chile–Korea	313,785	209,686	66.8%	175,853	207,342	33,833	16.1%
Oman–US	249,838	189,946	76.0%	169,290	189,490	20,656	10.9%
Canada–Chile	374,528	252,172	67.3%	208,606	248,393	43,566	17.3%
Total	**1,171,051**	**805,542**	**68.8%**	**683,488**	**797,811**	**122,054**	**15.2%**

DV = 1: the input can be used to make the output (main variable; it includes links that are possible but unconfirmed, marked uncertain). DV_strict counts only certain links (DV × (1 − uncertain)). DV_excl_remelt sets every direct-remelting HS2002 pair (route = recycle_direct, for example steel rod → steel ingots) to 0; a row coded on HS2002 that maps to several HS2002 pairs takes the highest value among them, so 3 Turkey–Singapore rows shown as recycle_direct keep 1 because another of their pairs has another route. The 591 Turkey–Singapore rows coded on their HS2012 codes are coded directly (none is recycle_direct).

3. Uncertain pairs

Table 2. Rows kept at DV = 1 with uncertain = 1, by reason

Reason	Turkey–Singapore	Nicaragua–Taiwan	Chile–Korea	Oman–US	Canada–Chile
The two AI coders (or the model and the textile rules) disagree	3,265	4,090	9,081	128	16,740
The two AI coders disagree: intermediate sibling or backward	75	83	561	0	1,109
Same-material intermediate of another heading, or another compound of one heading (30 Oct)	1,807	2,007	2,957	0	2,367
Remelting into another primary or semi-finished form of the same material (30 Oct)	2	861	994	0	1,893
Remelting excluded, but another specific route is possible (30 Oct)	3	1	19	0	46
Minor constituent of a garment or made-up: other fibre (not cotton or man-made) or other construction	0	10,196	16,839	16,863	16,471
Minor constituent of a yarn or fabric: other fibre (not cotton or man-made)	0	54	1,811	2,243	2,243
Open case in the ASEAN–Japan labels	191	1,162	1,555	1,422	2,693
HS concordance: the HS2002 pairs of a record differ	198	0	12	0	0
Possible feed (crustaceans → rock lobster)	0	4	4	0	4
Total uncertain	**5,541**	**18,458**	**33,833**	**20,656**	**43,566**

Most rows of these agreements are copies of PTA 1–5, so their uncertain pairs are those of PTA 1–5: mainly minor constituents of another fibre in textile goods, where US-style yarn-forward rules list almost every yarn and fabric as an input of each garment. The newly coded pairs add two kinds: pairs on which the two AI coders disagree (about half of them chemicals; the rest mainly stone and glass, mineral products, machinery and instruments), and pairs that the conventions of 30 October code as 1 with uncertainty (same-material intermediates of another heading, different compounds of one heading, and remelting into another form of the same material). The share of newly coded rows is Turkey–Singapore 40.5%, Nicaragua–Taiwan 7.2%, Chile–Korea 7.8%, Oman–US 0.0%, Canada–Chile 10.2%. Turkey–Singapore has the highest uncertain share because it has by far the highest share of newly coded rows, more than half of them chemicals.

4. How the codes were obtained

Table 3. Rows by coding method

Method	Turkey–Singapore	Nicaragua–Taiwan	Chile–Korea	Oman–US	Canada–Chile	Unique pairs
Copied from the ASEAN–Japan labels (step 7a)	2,567	68,746	78,749	51,310	93,740	349,143 (both)
Copied from the PTA 2–5 codes (step 7a)	10,457	127,059	210,702	198,528	242,440	
Two AI coders (steps 2–6)	8,254	14,511	22,836	0	36,961	53,849
Two AI coders on HS2012 codes (step 1)	591	0	0	0	0	589
Textile rules (step 7e)	0	705	1,492	0	1,383	2,006
Self-pairs (step 2)	4	6	6	0	4	6
Total	**21,873**	**211,027**	**313,785**	**249,838**	**374,528**	**405,593**

Copies come from the coded master of PTA 1–5: the ASEAN–Japan labels and the codes of Brunei–Japan, Australia–US, CAFTA and Bahrain–US (delivered on 19 October). The AI coders coded 53,696 pairs in 4,751 heading-pair units and a blind sample of 153 textile pairs. The model bands were not used for new pairs.

The conventions of 28–30 October (numbered as in section I.9 of the coders' rules, *Coding_Rules_2026_10_30.md*) were applied by rule to the newly coded pairs, including pairs both AI coders had coded 0 where a convention covers a whole class (remelting, metal articles of the same metal):

- Remelting (convention 3): 2,290 pairs from 1 to 0 (made into a finished product, vulcanised rubber, different or non-meltable polymers); 52 more stay 1 with uncertainty because another specific route exists. 566 pairs are certain 1s (back to the input's own upstream form) and 1,899 are 1 with uncertainty (another primary or semi-finished form), 586 of them coded 0 by both AI coders.
- Same-material intermediates (convention 2): 207 metal-article pairs from 0 to 1; parts, metal articles and compounds of one heading linked only by their material are 1 with uncertainty.
- Waste (convention 5): yarn, fabric or fibre → worn clothing (6309) and fibre → rags (6310) from 0 to 1 (textile rules); swine meat, offal and fat → pig bristles 0.
- Screen of one-sided 1s (section 5): 200 more pairs to 0 (the screen's other 10 pairs, pork → pig bristles, are counted under waste). Pairs both independent auditors coded 0 in the classes where Professor Zhang said so: 6 pairs to 0.
- Chapter 59 (textile rules): wicks and hosepiping → belting 0; coated or rubberised fabric → belting 1; gum-coated fabric (5901) → tyre cord fabric 1.

5. Reliability

- **Consistency with PTA 1–5.** The 349,143 copied pairs carry exactly the PTA 1–5 codes (0 differences in DV, uncertain and route).
- **Consistency across the five agreements.** 342,891 pairs (master_id, the HS2002 pair a row is coded under) appear in two or more agreements; DV is identical for all of them (0 differences). Uncertain, route and uncertain reason are identical too, except for 188 pairs that also appear in a row mapped to several HS2002 pairs (Turkey–Singapore HS2012 codes, invalid Chile–Korea codes), where the row takes the

highest DV of its pairs. Counting every HS2002 pair such a row maps to, 344,310 pairs appear in two or more agreements.

- **Two AI coders.** Claude and Codex coded 53,696 new pairs independently. They agreed on 57.7% (Cohen's kappa 0.28). Codex coded 1 far more often (39,553 pairs against 17,367): in 22,449 pairs Codex coded 1 and Claude 0, in 263 the reverse. Agreement was high for animal products (kappa 0.84), plant products (0.65) and textiles (0.61), and close to none for chemicals (0.03), food preparations (0.03) and stone and glass (0.07). Disagreements stay at DV = 1 with uncertain = 1 unless a convention settles them. On the blind textile sample (153 pairs) kappa was 0.80; on the 589 HS2012 pairs, 0.69.
- **Screen of one-sided 1s.** 16,508 pairs were coded 1 by one coder and 0 by the other and were not settled by a convention. Claude found 7,683 of them clearly invalid; Codex, blind, agreed on 210 (218 rows): tobacco → garments (180 pairs), pork → pig bristles (10), and 20 others such as coal → lignite. Only these became 0; the pork → pig bristles pairs are now coded by the waste convention, so the screen's own policy mark covers 200 pairs. Codex named a specific role for almost every chemical, which the convention on process inputs codes as 1 with uncertainty, so the screen is conservative.
- **Independent AI audit.** A stratified sample of 305 pairs (by how the code was reached: AI-coded pairs by input section, textile-rule pairs by rule, HS2012 pairs, and copies by origin) was coded by two new AI sessions (Claude and Codex) that saw only the rules and the pairs. They agree with each other on 87.2% of pairs (kappa 0.74). August reviewed the 57 pairs where the auditors disagree with each other (39) or both disagree with the final code (18). The true values of 16 sampled pairs were updated: 15 by the conventions of 30 October (8 of them Professor Zhang's own examples) and 1 by the rule on finished accessories (scarf → jacket = 0).

Table 4. Accuracy of the final codes (305 audited pairs, weighted by stratum size; 95% intervals)

	By pairs	By rows
DV, final (31 October)	96.5% (90.5–98.8)	98.8% (92.9–99.8)
DV, draft of 28 October, same true values	95.1%	98.0%
DV_strict, final	83.6% (74.7–89.8)	87.4% (77.5–93.3)

DV by rows	Turkey–Singapore	Nicaragua–Taiwan	Chile–Korea	Oman–US	Canada–Chile
Final	97.8%	98.6%	98.7%	99.1%	98.7%

14 sampled pairs remain wrong in DV: 5 complete machines built into another good that both AI coders coded 0 (convention 4 has not been applied to pairs coded 0), 6 one-sided 1s the screen kept, 1 other HS2012 pair, and 2 copies of PTA 1–5, which are not re-judged. Six sampled pairs were set to 0 directly because both auditors coded them 0; counting them as errors gives 96.0% by pairs and 98.5% by rows. DV_strict is lower because many pairs whose true value is 1 are coded 1 with uncertainty. This is an AI audit reviewed by a research assistant, not an expert hand-coding.

6. Revisions of 28–31 October 2026

Table 5. DV = 1 and uncertain rows by version

Agreement	DV = 1, 28 Oct	DV = 1, final	DV_strict = 1, 28 Oct	DV_strict = 1, final	Uncertain, 28 Oct	Uncertain, final
Turkey–Singapore	15,034	14,559	10,823	9,018	4,211	5,541
Nicaragua–Taiwan	138,629	139,179	122,385	120,721	16,244	18,458
Chile–Korea	209,768	209,686	178,437	175,853	31,331	33,833
Oman–US	189,946	189,946	169,290	169,290	20,656	20,656
Canada–Chile	253,091	252,172	210,258	208,606	42,833	43,566
Total	**806,468**	**805,542**	**691,193**	**683,488**	**115,275**	**122,054**

- **28 October (draft):** coded under the conventions of 19 October. The Turkey–Singapore draft also had 2,273 rows with input “WO” or “any” and no code; they are excluded from the figures above.
- **31 October (final):** the decisions of 28 October (screen of one-sided 1s, deletion of “WO” and “any”, chapter 59) and Professor Zhang’s questionnaire answers of 30 October, applied to the newly coded pairs. Against the draft, 1,209 HS2002 pairs went from 0 to 1, 2,509 from 1 to 0, and 5,174 changed only their uncertain flag (HS2012 pairs: 4, 3 and 49). The share of rows with DV = 1 moved from 68.9% to 68.8%; DV_strict from 59.0% to 58.4%, mainly because convention 2 turned 4,757 pairs (8,662 rows) of certain 1s into 1s with uncertainty. Oman–US is unchanged because all its pairs are copies of PTA 1–5.

7. Notes

- **Scope.** Codes delivered for PTA 1–5 are not re-judged (Professor Zhang, 30 October), and pairs copied from them keep their codes: 92.6% of the rows here. The conventions of 30 October therefore change only the newly coded pairs.
- **Applied in part.** Generic process-input claims became 0 only through the screen of one-sided 1s (and, for sampled pairs both auditors coded 0, through the exception in section 5). The conventions that could raise a 0 to 1 (complete machines, cut or refitted goods, associated minerals recovered from an ore, crushed goods as filler, waste of a component) were not applied to pairs both coders had coded 0. Distinctions between a certain 1 and a 1 with uncertainty were not re-judged pair by pair; they apply to future coding and affect only DV_strict.
- **To be confirmed by Professor Zhang.** DV as the main variable, with DV_strict as a robustness check. Whether the complete machines built into another good that both AI coders coded 0 (6,733 HS2002 pairs) are re-reviewed under convention 4. Three extensions of his answers: thermosets and other non-meltable plastics are 0; rubbers in primary forms only to the same polymer; no laminate route between different plastics (“other plastics” also count as a polymer of their own). Also: which convention applies where metal articles (convention 2) and remelting into a finished product (convention 3) overlap (convention 2 for now).
- concord_flag = 1 marks 5,235 Turkey–Singapore rows whose HS2012 code is not one-to-one with HS2002, and 614 Chile–Korea rows with an invalid code (610340, 610344, 610345), mapped to every HS2002 subheading of 6103.4. The HS concordance will follow the project’s conversion package once it is adopted.

- The AI coders are Claude (Anthropic) and Codex (OpenAI); their codes and one-sentence reasons are kept in dv_ai1, dv_ai2, why_ai1 and why_ai2 where they coded the pair.